QUANTITATIVE METHODS 1

Worksheet 6 – Index numbers

Detailed solutions

Exercise 1 – Detailed solution

$$I_{2024/2023} = 115, \qquad I_{2025/2024} = 110, \qquad I_{2025/2023} = 126.5.$$

Circularity:

$$\frac{115 \times 110}{100} = 126.5.$$

Reverse index:

$$I_{2023/2025} = 100\frac{80}{101.20} \approx 79.0514.$$

Reversibility:

$$126.5 \times 79.0514 \approx 10000.$$

Rebasing to 2024=100:

$$I_{2023}^{(2024)} \approx 86.96, \qquad I_{2024}^{(2024)} = 100, \qquad I_{2025}^{(2024)} = 110.$$

Index 126.5 means the 2025 level is 26.5% above 2023, not 126.5% higher.

Exercise 2 – Detailed solution

$$\sum p_0q_0 = 800, \quad \sum p_1q_0 = 980, \quad \sum p_0q_1 = 860, \quad \sum p_1q_1 = 1050.$$

Price indices:

$$I_P^L = 122.50, \qquad I_P^P \approx 122.09, \qquad I_P^F \approx 122.30.$$

Quantity indices:

$$I_Q^L = 107.50, \qquad I_Q^P \approx 107.14, \qquad I_Q^F \approx 107.32.$$

Value index:

$$I_V = 131.25.$$

Factor reversal:

$$\frac{I_P^F I_Q^F}{100} \approx 131.25 = I_V.$$

Exercise 3 – Detailed solution

$$I_{3/0}^{chain} = 100(1.05)(1.08)(0.97) = 109.998 \approx 110.$$

The final level is about 10% above the initial level despite the final 3% annual decrease. Successive changes compound multiplicatively.

Deflation:

$$\textit{Real value} = \frac{2.4}{120/100} = 2.0 \text{ million euros.}$$